

# Hysteresis Analysis

## Effect of MD1 Conditioning on SFTPRO & LHC Cycles

MBB max-speed + NMR campaign — 2026-03-06

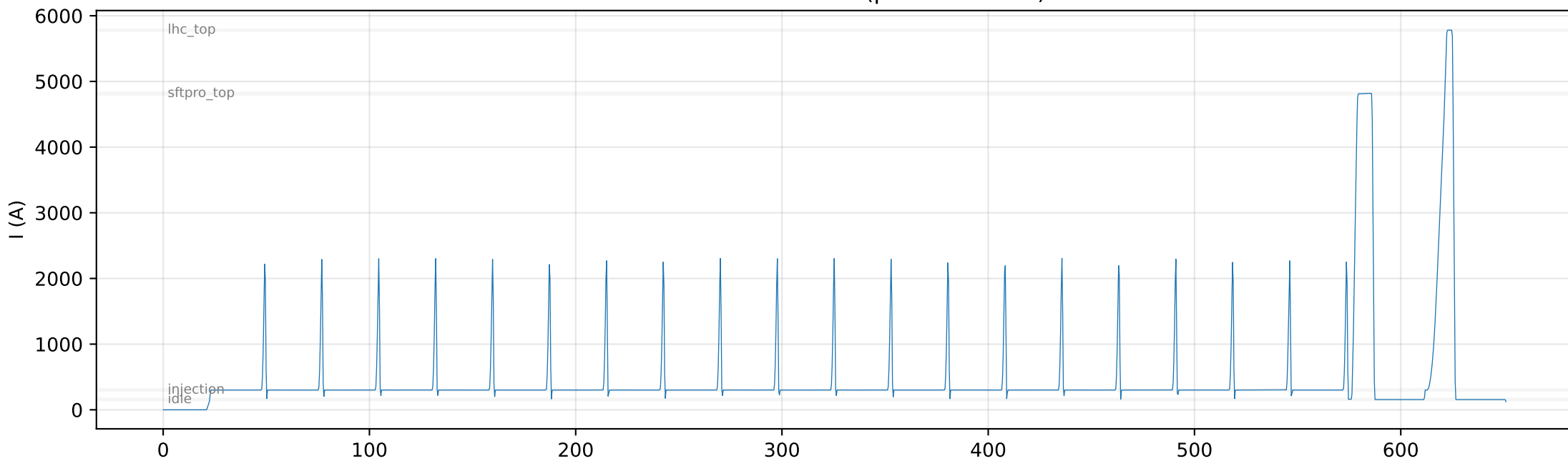
200 GeV session: 20× full MD1 (peak ~2267 A) → SFTPRO → LHC

26 GeV session: 20× flat MD1 (peak ~301 A) → SFTPRO → LHC

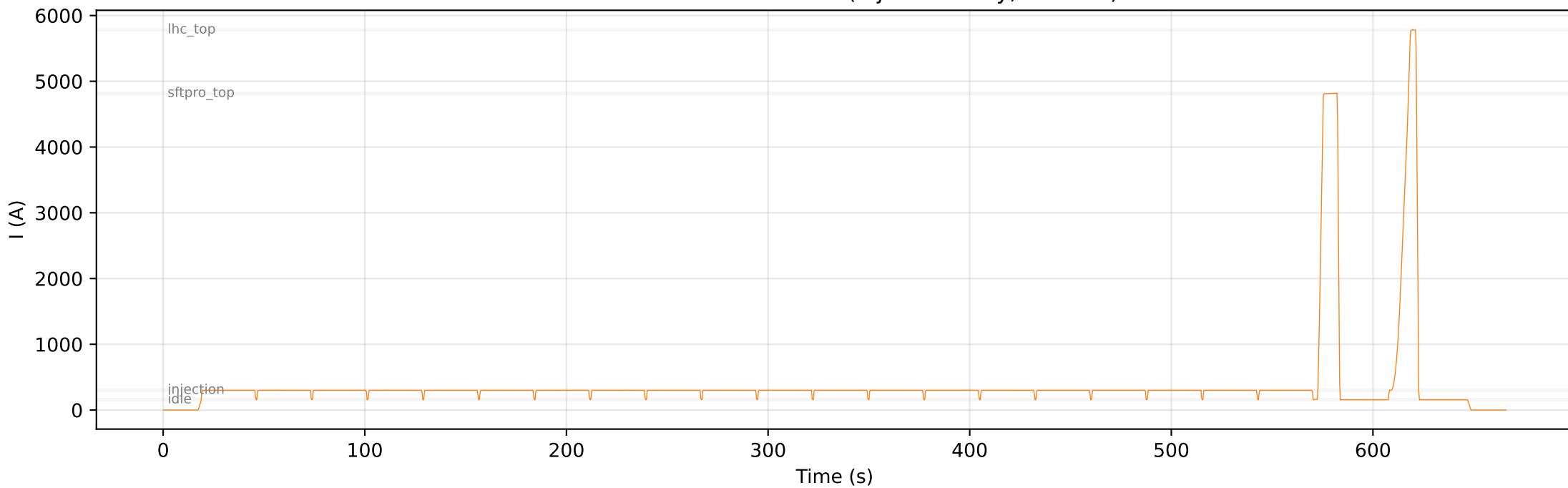
Body segment • Encoder offset corrected • Plateau by  $|\text{ramp rate}| < 5 \text{ A/s}$

# Current Profile — Both Sessions

## 200 GeV — Full MD1 (peak ~2267 A)

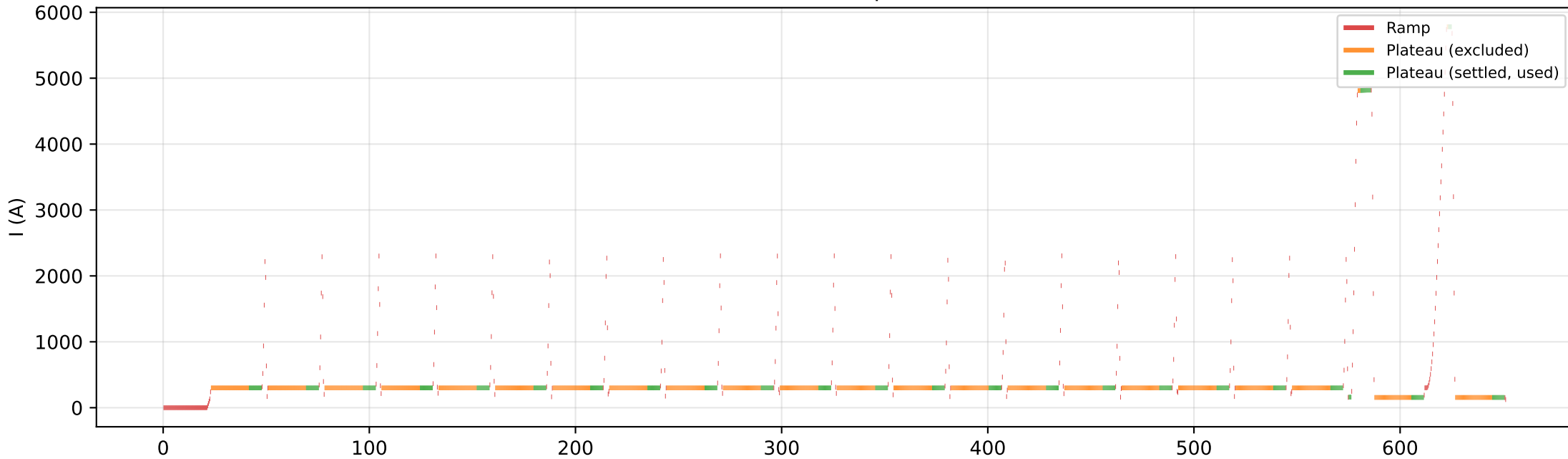


## 26 GeV — Flattened MD1 (injection only, ~301 A)

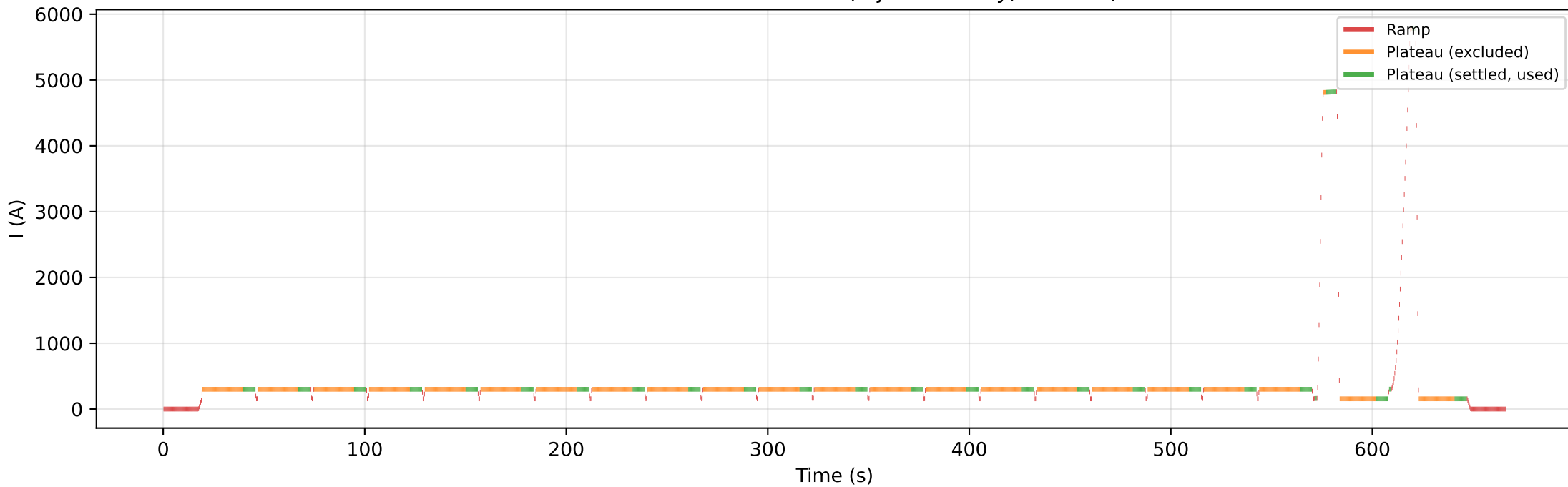


# Current Profile with Turn Classification

## 200 GeV — Full MD1 (peak ~2267 A)

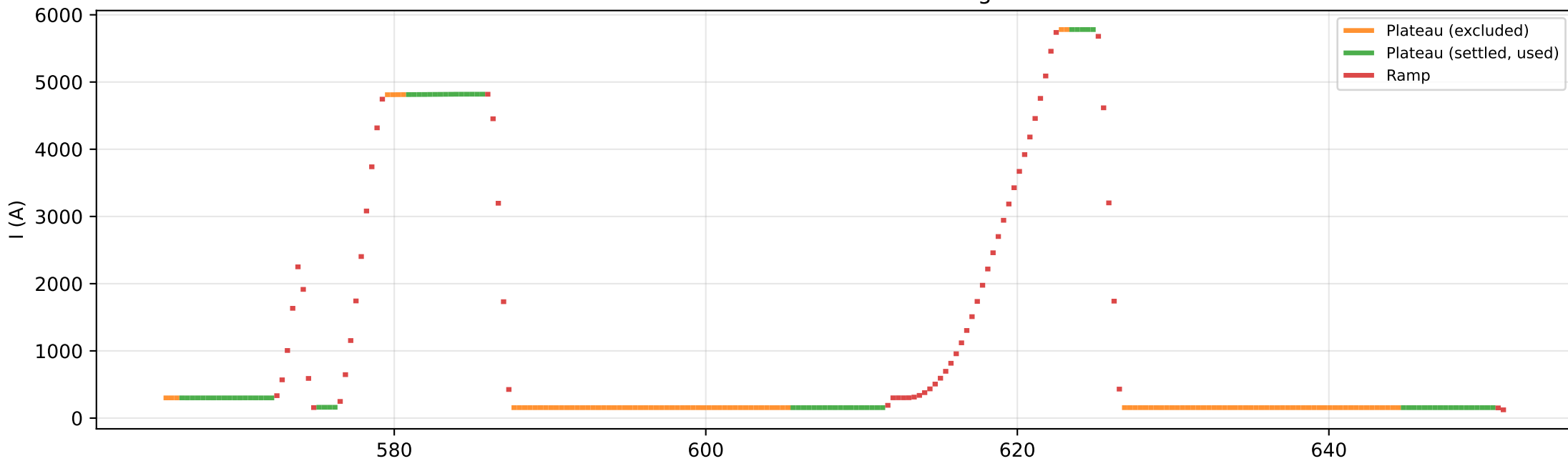


## 26 GeV — Flattened MD1 (injection only, ~301 A)

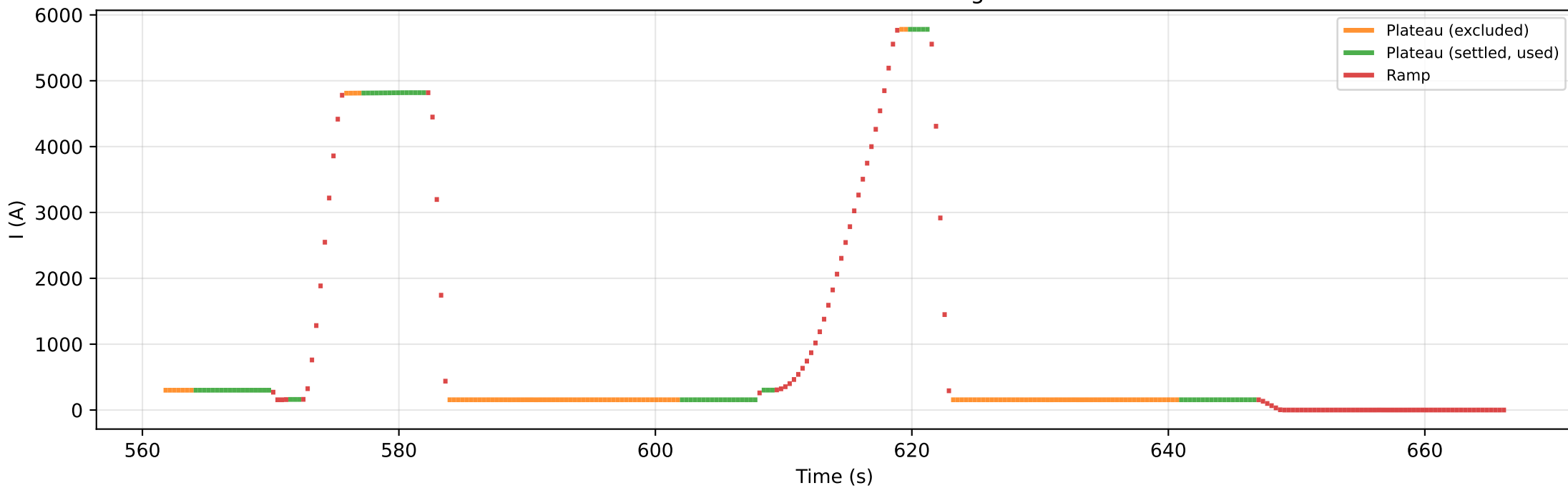


# SFTPRO + LHC Region — Turn Classification (zoom)

## 200 GeV — SFTPRO + LHC region

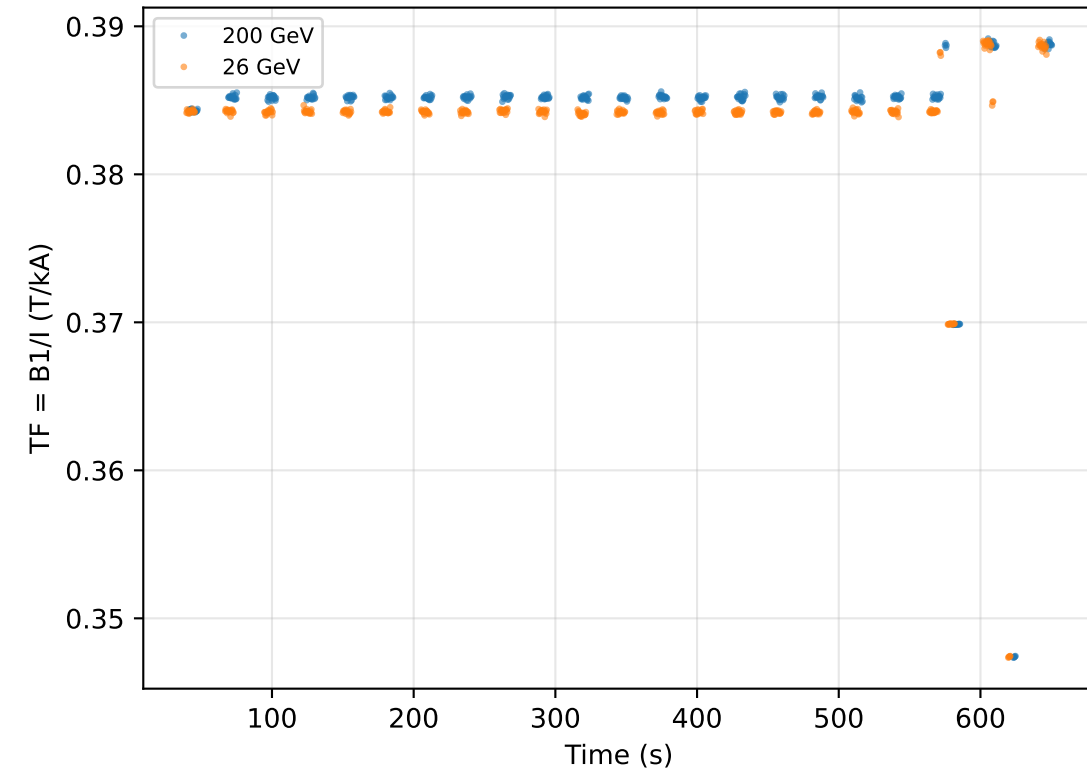


## 26 GeV — SFTPRO + LHC region

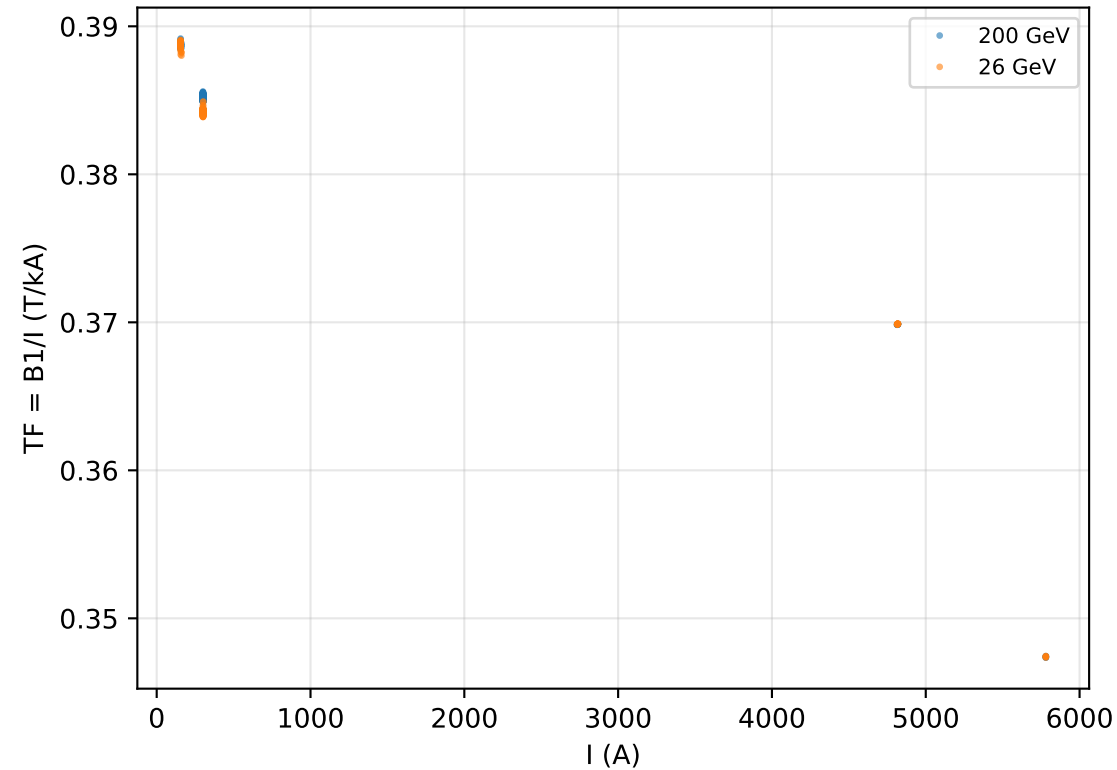


# Transfer Function — Settled Turns Only (body)

## Transfer Function vs Time

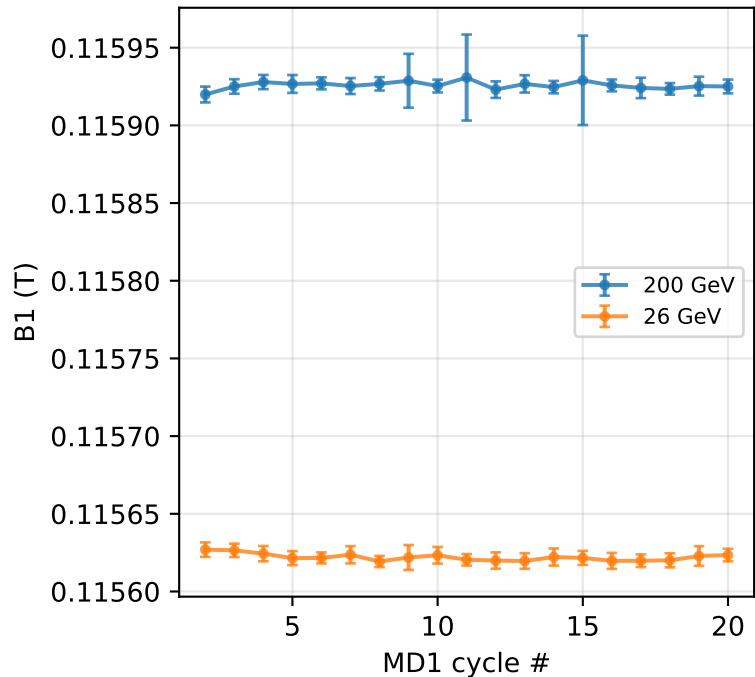


## Transfer Function vs Current

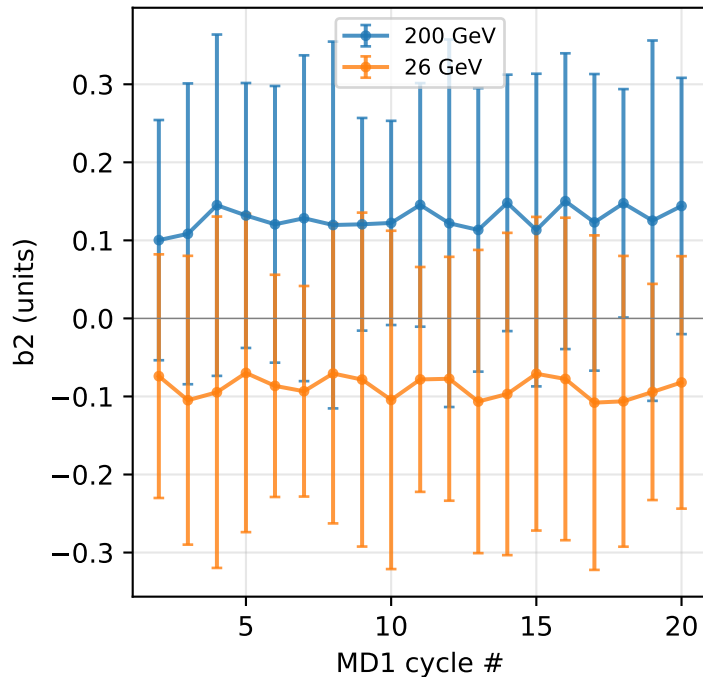


# MD1 Minor Loop Convergence — Injection Plateau (body, cycle 1 excluded)

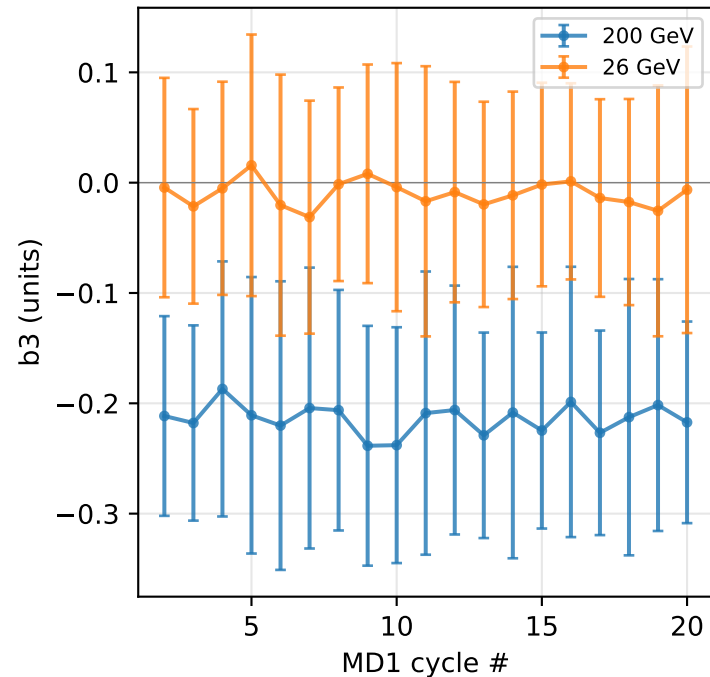
## B1 (T) at injection (~301 A)



## b2 (units) at injection (~301 A)

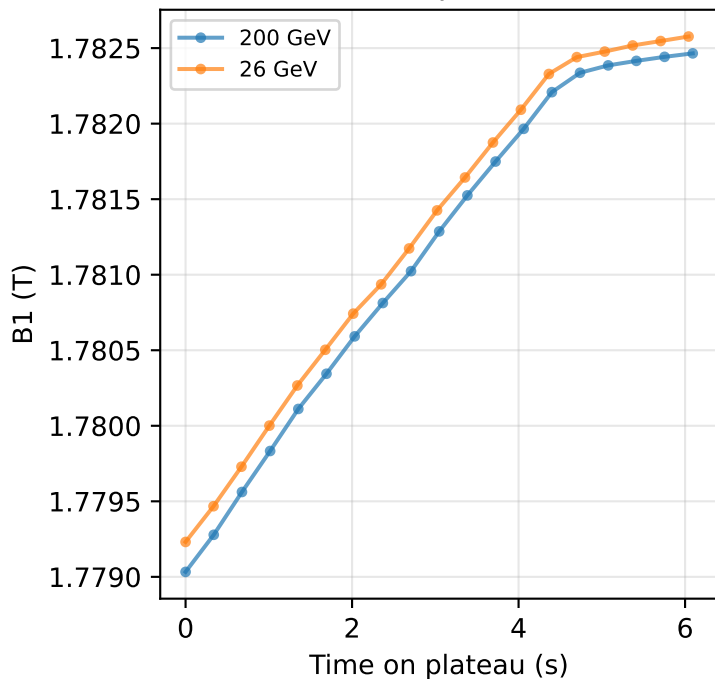


## b3 (units) at injection (~301 A)

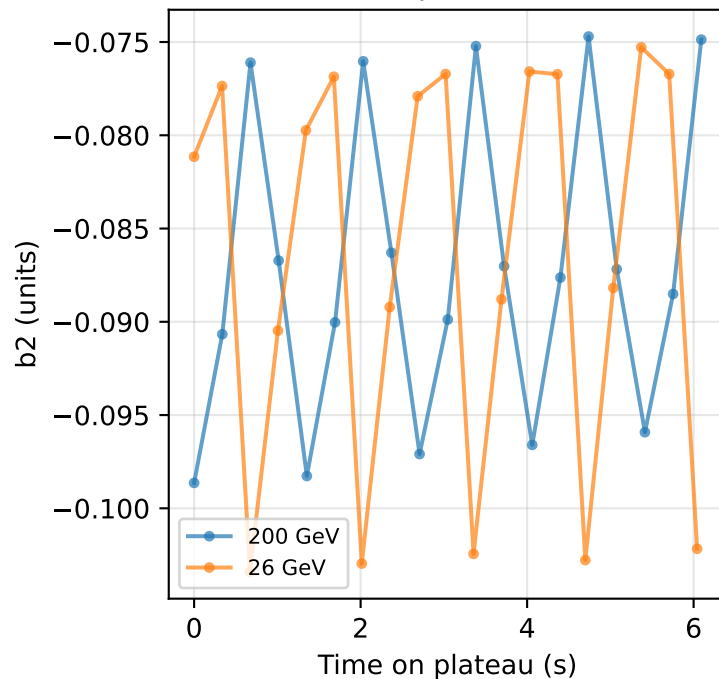


# SFTPRO Top Plateau — Turn-by-Turn (body)

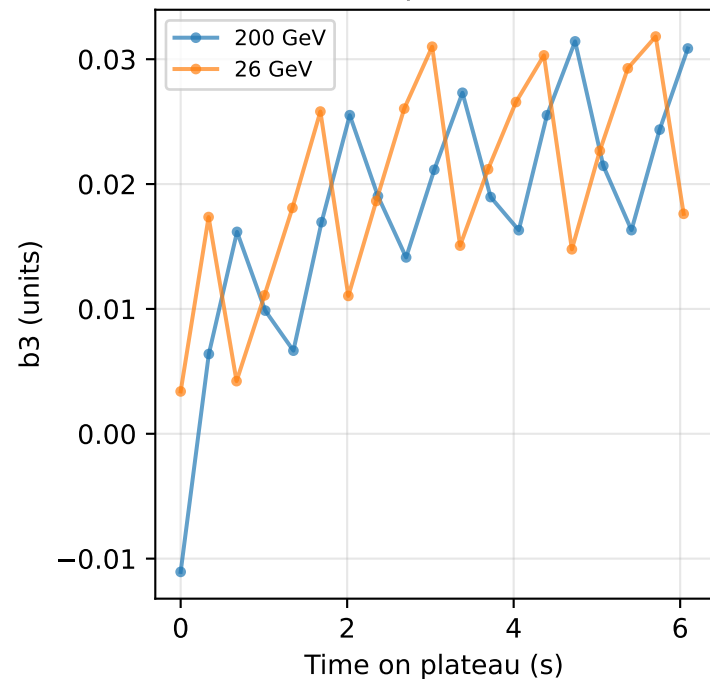
## SFTPRO top — B1 (T)



## SFTPRO top — b2 (units)

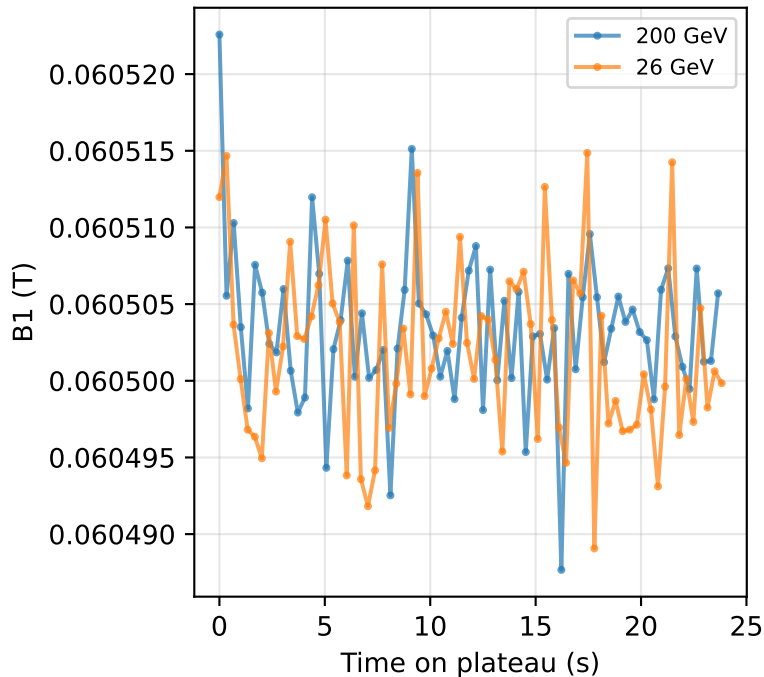


## SFTPRO top — b3 (units)

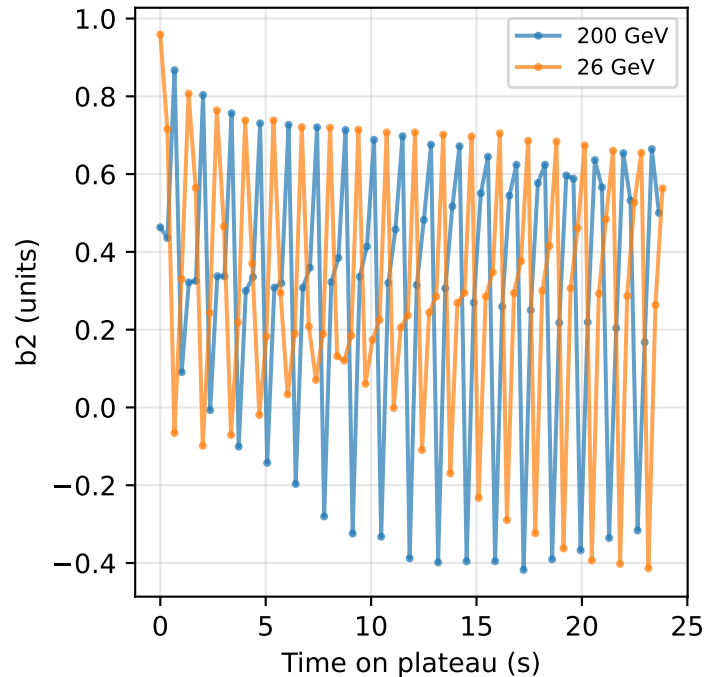


# Post-SFTPRO Idle Plateau — Turn-by-Turn (body)

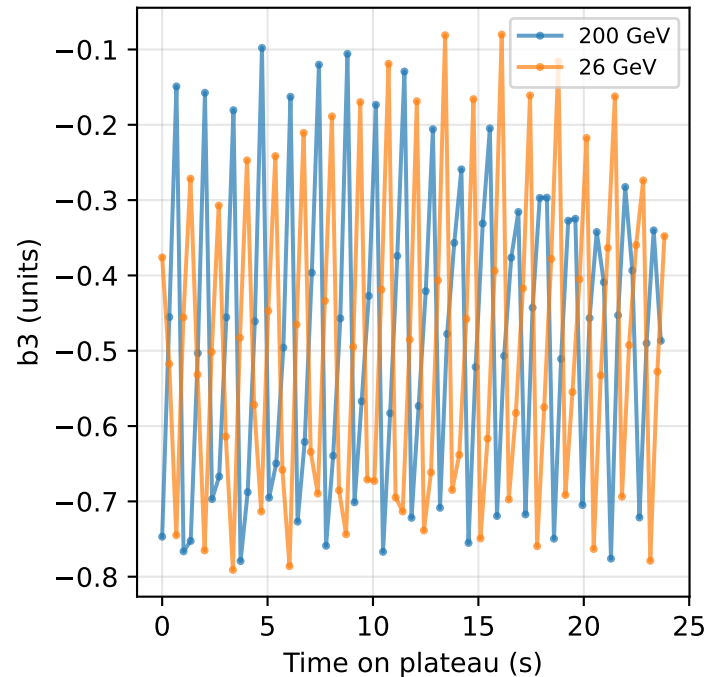
## Post-SFTPRO idle — B1 (T)



## Post-SFTPRO idle — b2 (units)



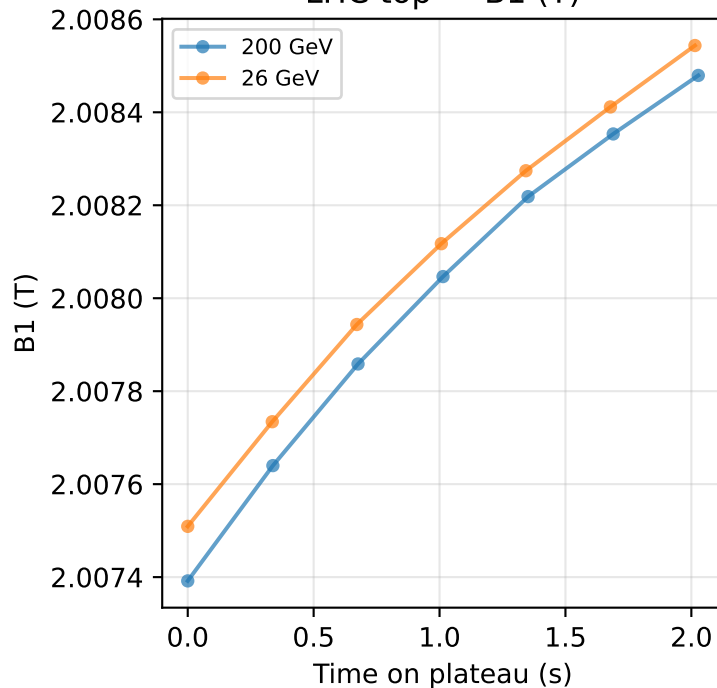
## Post-SFTPRO idle — b3 (units)



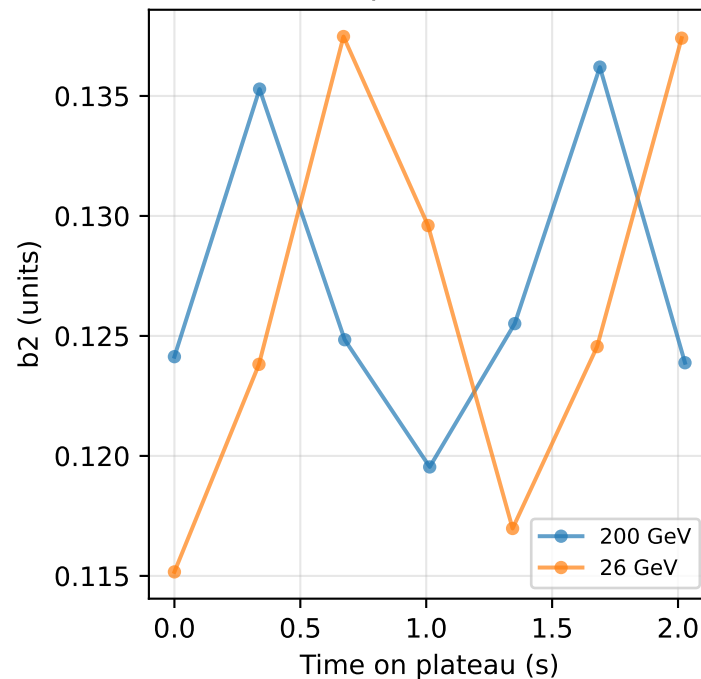


# LHC Top Plateau — Turn-by-Turn (body)

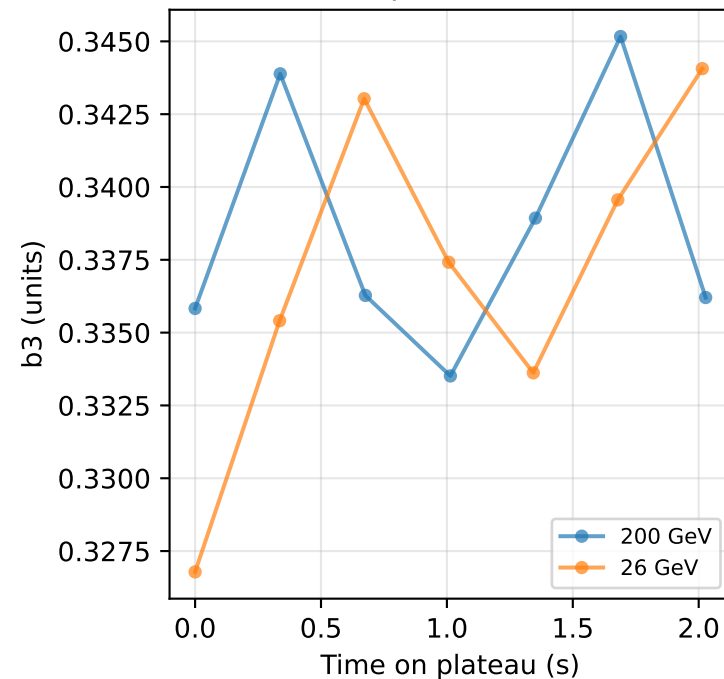
## LHC top — B1 (T)



## LHC top — b2 (units)

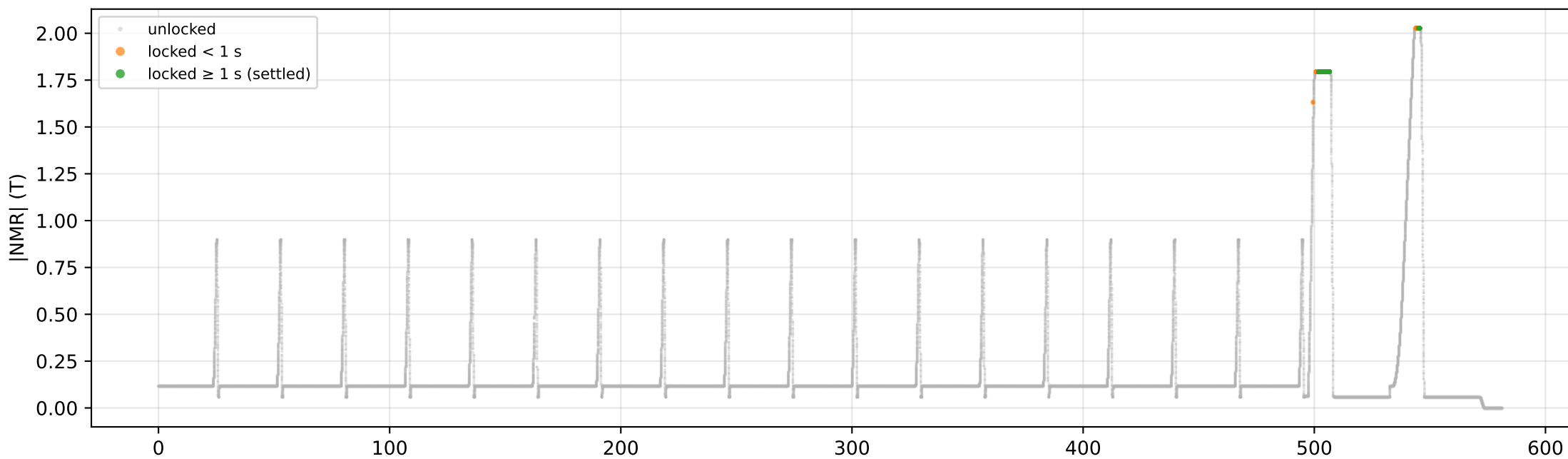


## LHC top — b3 (units)

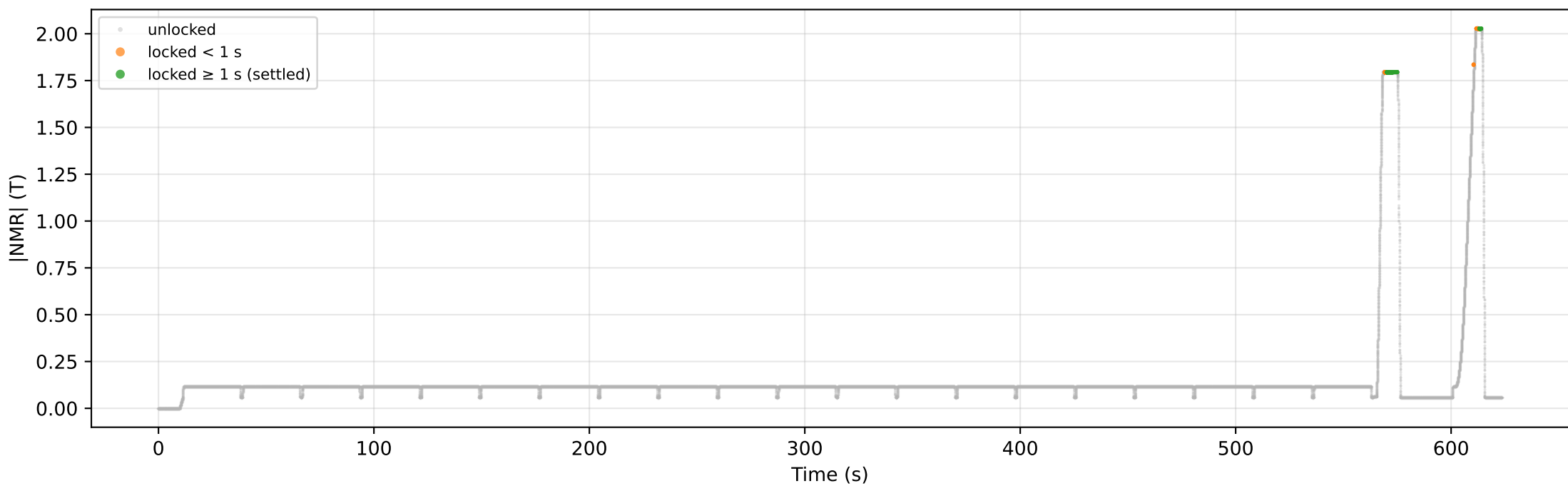


NMR Lock Status — Grey=unlocked, Orange=early, Green=settled

200 GeV — NMR

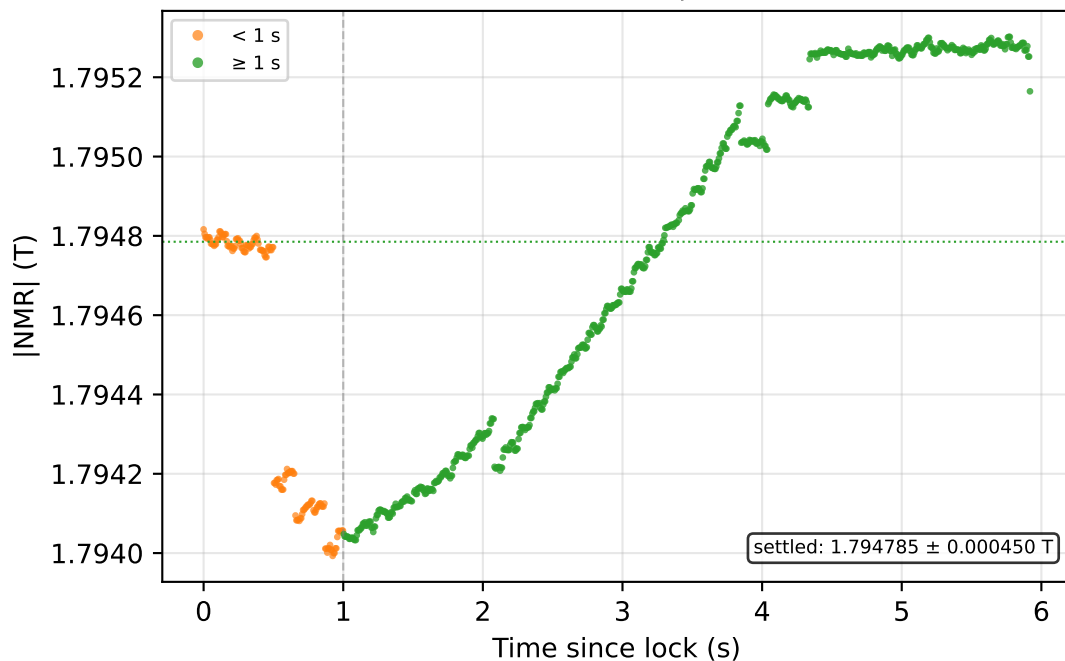


26 GeV — NMR

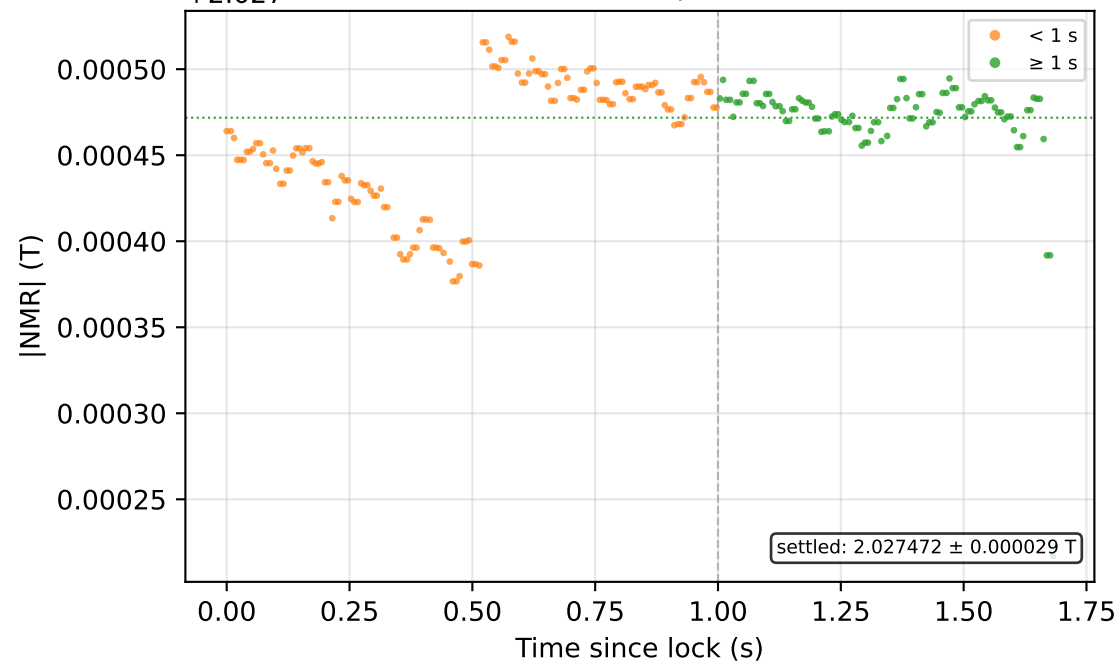


# NMR Lock Periods — Zoom

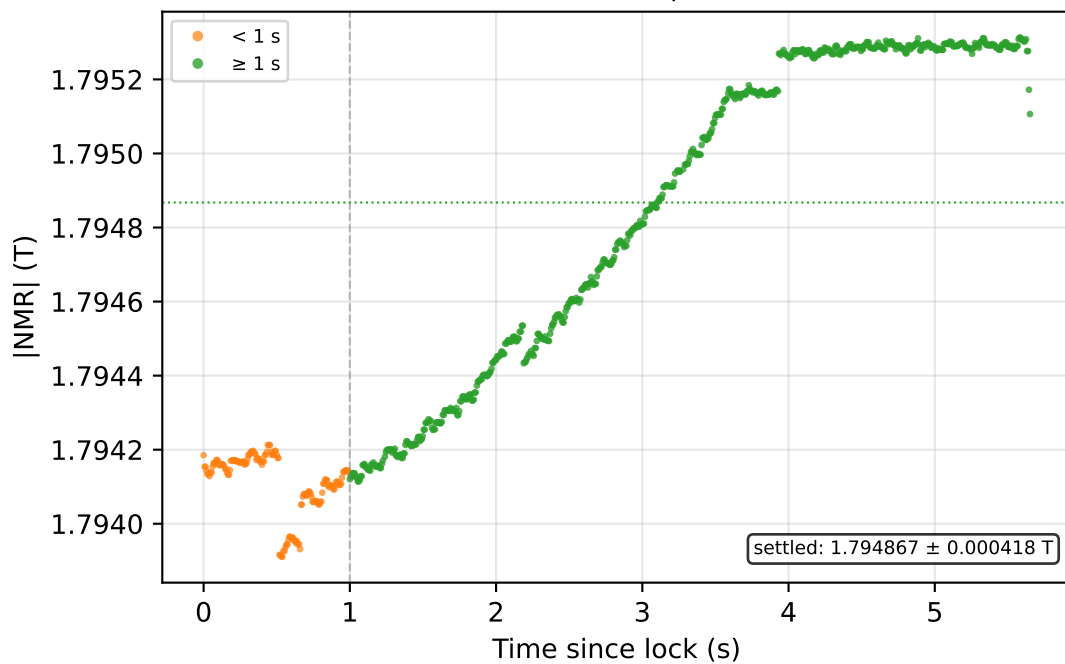
200 GeV — SFTPRO top (5.9 s lock)



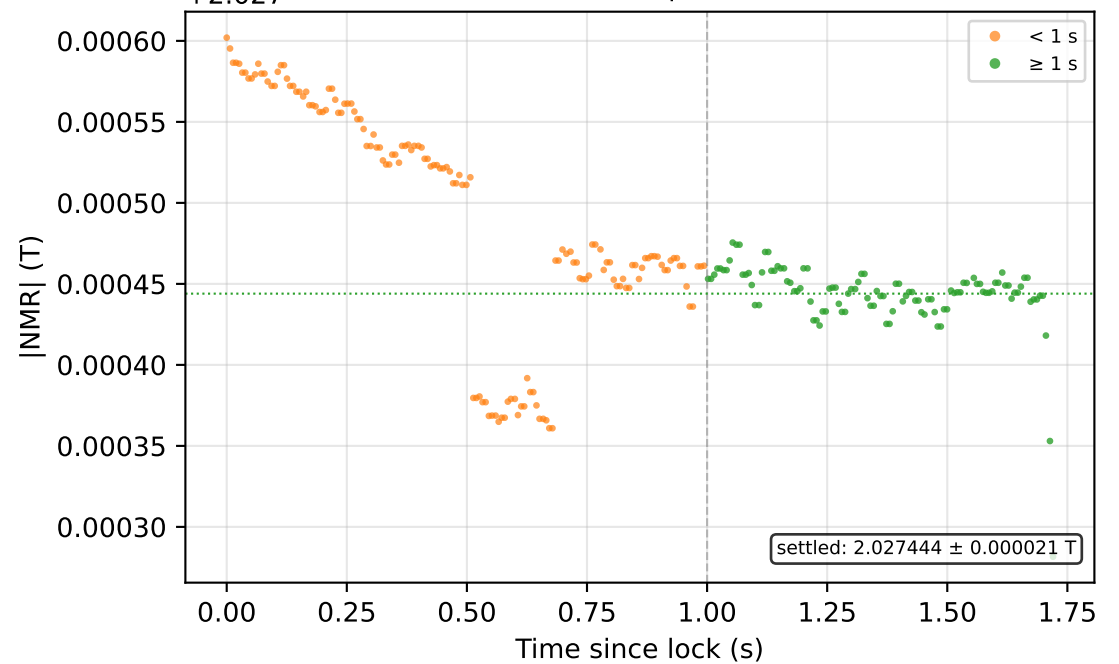
200 GeV — LHC top (1.7 s lock)



26 GeV — SFTPRO top (5.7 s lock)



26 GeV — LHC top (1.7 s lock)



# Hysteresis Summary: $\Delta = 26 \text{ GeV}$ (flat MD1) – $200 \text{ GeV}$ (full MD1)

| Plateau                   | 200 GeV<br>B1 (T) | 26 GeV<br>B1 (T) | $\Delta$ B1<br>( $\mu\text{T}$ ) | 200 GeV<br>b2 (units) | 26 GeV<br>b2 (units) | $\Delta$ b2<br>(units) | 200 GeV<br>b3 (units) | 26 GeV<br>b3 (units) | $\Delta$ b3<br>(units) |
|---------------------------|-------------------|------------------|----------------------------------|-----------------------|----------------------|------------------------|-----------------------|----------------------|------------------------|
| SFTPRO top<br>(4816 A)    | 1.781578          | 1.781703         | <b>125.5</b>                     | -0.087                | -0.086               | <b>0.001</b>           | 0.021                 | 0.023                | <b>0.002</b>           |
| SFTPRO→LHC idl<br>(155 A) | 0.060503          | 0.060500         | <b>-3.7</b>                      | 0.296                 | 0.278                | <b>-0.019</b>          | -0.465                | -0.457               | <b>0.007</b>           |
| LHC injection<br>(302 A)  | nan               | 0.116350         | <b>nan</b>                       | nan                   | -0.094               | <b>nan</b>             | nan                   | -0.084               | <b>nan</b>             |
| LHC top<br>(5781 A)       | 2.008191          | 2.008258         | <b>66.8</b>                      | 0.126                 | 0.129                | <b>0.003</b>           | 0.338                 | 0.340                | <b>0.002</b>           |
| Post-LHC idle<br>(155 A)  | 0.060499          | 0.060495         | <b>-4.3</b>                      | 0.233                 | 0.276                | <b>0.043</b>           | -0.472                | -0.459               | <b>0.013</b>           |
| SFTPRO top<br>(NMR)       | 1.794785          | 1.794867         | <b>82.3</b>                      | —                     | —                    | —                      | —                     | —                    | —                      |
| LHC top<br>(NMR)          | 2.027472          | 2.027444         | <b>-27.8</b>                     | —                     | —                    | —                      | —                     | —                    | —                      |

Body segment only • Encoder offset corrected (equiv. to  $-\pi/2$ ) • Plateau:  $|\text{ramp rate}| < 5 \text{ A/s}$   
NMR: settled values  $\geq 1 \text{ s}$  after lock acquisition

# Static Hysteresis Impact — Bar Chart

